Russians launch IC sales campaign with ambitious exhibit at Paris Salon

Soviet circuits match performance of many off-the-shelf American IC's; but unusual packaging may scare off potential buyers in Western Europe

Native semiconductor producers in Western Europe, already hard put to hold their own against hard-driving U.S. competitors, now face an added challenge—from the Soviet Union.

The Russians hastily started their West European sales campaign at the Paris Components Salon earlier this month, where their display was mobbed by showgoers itching to see Soviet integrated circuits. Spurred by the success of their last-minute Paris exhibit—their request for space was filed only two weeks before opening day and their stand had to be squeezed into a bar—the Soviets now plan to mount a bigger display at next month's International Electronic Components Show in London. Then they will follow up with a mobile exhibit that will swing through Scandinavia later this year.

Standouts among these Soviet exhibits are 12 families of IC's—both hybrids and monolithics—plus four families of MOS logic circuits. At Paris, Western experts rated them as state-of-the-art. "We thought they were much further behind," said Roger Combes, marketing director of SGS-France, a subsidiary of Italy's Società Generale Semiconduttori. "They're practically as advanced as the U.S."

Well placed. And the Russian wares, indeed, could find a place—although not at the top of the line—in the catalog of most U.S. semiconductor producers. The Soviet transistor-transistor logic, for example, has a delay time of 40 nanoseconds and a power dissipation of 18 milliwatts. That compares with 40-nsec delay and 60-mw dissipation for Texas Instruments' series 54/74 flip-flop. And the Russians' emitter-coupled logic, with 3-nsec delay and 80-mw dissipation, can stand up to Motorola's MECL II circuits.

With nothing revolutionary to sell, the Russians could be tempted to win their place in the market by undercutting prices as other newcomers have done. This they will not do, insists Nicolai Gssaev, deputy chief of the Soviet electronics export agency, Machpriborintorg. "Our prices will be competitive," he says, "and we're aiming at higher quality than Western producers offer." But Gssaev frankly admits the goal still has to be reached.

Prospects. Even so, the Russians figure they have solid prospects in Western Europe. Most likely first big customer: Italy's Ing. C. Olivetti & Co., which may buy Russian-made MOS for desk calculators.

And a gaggle of European firms have lined up at Machpriborintorg in quest of distribution rights for Soviet IC's. The Soviets, though, plan to market IC's directly to equipment manufacturers through their own men working out of embassies in Western Europe. For discrete components, on the other hand, there'll be capitalist middlemen. Already a West German distributor called Trans-Electronics Alfred Hempel KG has started peddling the Machpriborintorg line—IC's excepted—in its home market.

Hip, hip, array! Alexander Vasenkov, research chief of the Russian electronics-industry ministry, hinted at Paris Components Show that LSI's for export were in the offing.

No matter who's selling for them, the Soviets figure to run into higher-than-usual sales resistance.

For one thing, there's the question of delivery. Many Western observers are skeptical about Russian claims that they can deliver immediately anything in their limited IC product line. The Russians themselves say that there'll be only 150,000 IC's for export this year but maintain their target for 1970 is "several million."

For another, there's a problem in

packages. Russian encapsulation and pin-spacing differ from the norms in Western Europe. Thus there's a double drawback for users of the Soviet IC's: an added design difficulty and—much worse—a single-source supply. Because the Russian packaging is so different, there would be no backup source available in Western Europe.

No threat yet. Above all, the Russians will be competing against the tough and seasoned sales troops of U.S. producers like TI, Motorola, and Fairchild. Until the Soviets have an edge in technology or in prices, they don't pose a major threat in IC markets, most European semiconductor men feel.

But the Russians could nail down a substantial market in semiconductor raw materials, most of these same observers think. Supplies of high-quality silicon are tight in Europe, and the Russians say they have unlimited quantities to sell. Among the Soviet offerings are wafers with diameters of 75 millimeters, half again that of the wafers normally made in the U.S. and Europe. In addition to silicon, the Russians say they have available for immediate export silicon-on-sapphire wafers.

Already the Russians have one hot prospect for wafers. SGS-France intends to buy 50-mm wafers from them if the price is right.

More to come. Unless they get absolutely nowhere with their initial effort in IC's, the Russians apparently will plump out their line of advanced products. Alexander A. Vasenkov, chief of the research department at the electronics-industry ministry in Moscow, hints there's much more to come. "Back in Russia," he says, "we have both MOS and bipolar LSI arrays, plus lots of other interesting things. This is just our first step."

And as a tipoff to what might be in the works, the Russians at Paris had a prototype television set with all its functions—power supply, tuner, and audio output excepted—handled by a dozen hybrid IC's. The Russians have no plans to produce the set; but it shows, Vasenkov points out, "what we're doing to learn the possibilities of integrated circuits."

France

Pretty fair exchange

Ask data-processing consultants what the handiest computer terminal is and they'll quite likely launch off into a long debate over the relative merits of printers, punchers, and cathode-ray tubes. Ask the men at IBM's La Gaude research facility in France, and the unanimous reply will be a push-button telephone.

The reason for their concurrence is a new piece of IBM hardware, an electronic telephone exchange that can link 750 push-button telephones to a central computer. Each phone keyboard, then, becomes an input terminal for the computer; and the receiver can serve as an output terminal for voice readout. The exchange can also tie the computer to more conventional terminals like typewriters. And, of course, it switches ordinary phone calls.

With its new exchange, the 2750, IBM is offering its European customers a service U.S. customers can't get without having to turn to special lines and modems. Company officials won't talk about plans to get the 2750 exchange onto the American market, but it's a safe bet that IBM is itching to do just that.

Closed shop. Trouble is, the telephone-hardware market is still the fief of the American Telephone & Telegraph Co. And although the Federal Communications Commission has forced AT&T to ease its stance against non-Bell hardware, the Justice Department isn't likely to see IBM as the ideal candidate for a share of Bell's monopoly.

For the moment, IBM will concentrate on sales in Europe, where the government-run telephone systems in France, West Germany, and Italy have tentatively approved the exchange. Monthly rents for the 2750 range from $1,800 to $12,000, depending on the number of lines and optional extras.

Like Bell's 101ESS private exchange, IBM's 2750 features semiconductor switching rather than the electromechanical cross-points that characterize the first generation of "electronic" exchanges. But where Bell opted for transistors, IBM has selected thyristors. They switch in about 1 microsecond, making them directly compatible with the logic circuits of the control unit, which is paired with a 512,000-bit ferrite-core memory.

The exchange can be programed for a variety of special telephone services—paging, abbreviated dialing, and the like.

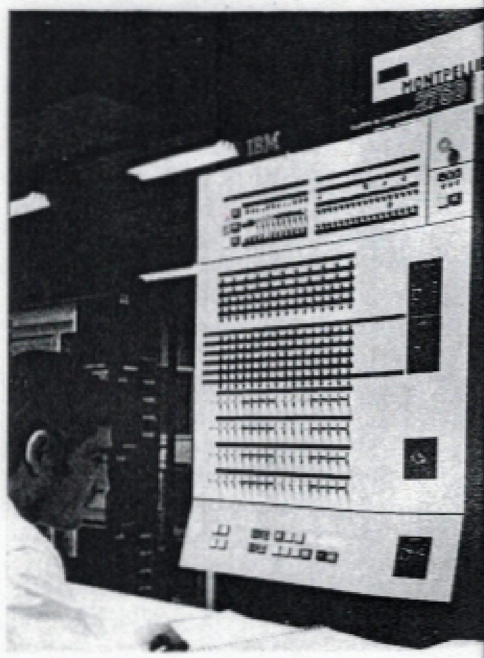
Great Britain

Graphic description

Computer graphics, at first glance, seem ideal for circuit designers. A little mistake? No problem. Simply erase with a light pen, redraw, and the computer does the rest.

It does it at a price, though, and the price has often been more than the traffic will bear, particularly when the data for the graphics is moving in and out of a big, time-shared computer. As a result, graphics design aids haven't caught on as strongly as most thought they would.

A way out may be offered by a small satellite computer that takes care of the graphics for a big com-



Quick switch. IBM's electronic exchange turns push-button telephones into computer input terminals.

Light writing. Heinz Lemke of the Cambridge University Mathematical Laboratory developed the command scheme for the Pixie CAD system. Only the light pen—there are no buttons to push—is used to change designs.

puter that handles the analysis of circuits after the designers have drawn them in near-final form. In the U.S., Bell Labs tried the idea and apparently found it wanting. In Britain, though, three researchers at Cambridge University have made it work in a system they call Pixie.

Unbuttoned. The three—Neil Wiseman, Heinz Lemke, and John Hiles—maintain that Pixie's economics are better than anything they've heard about so far. Another boost for graphics they see in Pixie is its ease of use—all inputs are by light pen; there are no pushbuttons to fiddle with. The first Pixie system, in fact, has turned out so well that the trio talked about plans for a more sophisticated version—Pixie 2—at the mid-April CADEX computer-aided-design conference at Southampton University.

Pixie uses as its main computer an ICT Atlas Mark 2 machine with a 120,000-word, 48-bit core memory and a multi-access system developed at Cambridge. The satellite computer is a Digital Equipment Corp. PDP 7 with an 8,000-word, 18-bit store; it's paired with a Digital Equipment 340 display.

Roughly speaking, the PDP 7 handles the interactive computing and the big computer is tapped only for analysis of designs. More important, the satellite machine holds off making changes in its stored data structure until the designer is satisfied with his schematic. At this point, he instructs the computer to feed the contents of a temporary display file into an up-compiler whose routines change the data structure.

This changed data structure is transformed into a permanent display file and the designer can then manipulate his preliminary design by shifting from "drawing" to "pointing." In the pointing mode, circuit-element indicators stored in the permanent file blink when the pen is aimed at them. They can be erased, moved around the display, or assigned a value. In Pixie 1, there are three levels of manipulation—individual circuit elements, nodes of several elements, and entire subcircuits. In Pixie 2, there'll be eight levels.

Japan

Diodes galore

Prospects for massive optical memories, many experts say, depend largely on the matrixes needed to read out data stored as interference patterns on holograms.

If the experts are right, prospects have never looked better for holographic memories. A research team at the Semiconductor Research Institute in Sendai, Japan, has found a way to pack upwards of 40,000 interconnected photodiodes onto an 8-by-8-millimeter section of a silicon chip.

That's the equivalent of more than 60,000 diodes per square centimeter. And Junichi Nishizawa, who heads the Sendai team, says this is by far the highest density yet achieved in a photodiode matrix. Quite likely so. A matrix man at Bell Labs, whose work in this field is world class, terms the Sendai development "very respectable." Bell, though, is trying for large arrays rather than high-density ones.

Nishizawa and his men have been fabricating their 201-by-201 arrays on very thin wafers—about 30 microns, or only a third the thickness of wafers normally used for integrated circuits. Their extreme thinness makes the wafers extremely tough to handle, and yields are very low. But because the diodes are on a 40-micron pitch and because the diode structures have to go all the way through the wafer, it's essential to keep the wafer thin.

One . . . The matrixes start as p-type silicon wafers with silicon dioxide passivation on both sides. The first step toward a matrix is to put in a grid of front-to-back connections through the wafer. This step starts on the back side of the wafer, where moats 10 microns wide and between 5 and 10 microns deep are etched in. This gives the back side a striped look, with the stripes on a 40-micron pitch.

Then a grid of 10-micron windows, again on a 40-micron pitch, is opened through the silicon dioxide on the front side. The grid is aligned so that the windows lie directly above the moats on the back side. With both sides prepared, n-type impurities are diffused into the wafer from both sides until they meet in the center to form a through connection for each diode.

Two . . . In the second step, the counterelectrodes and isolation regions that surround each diode are formed. The oxide coating on the front side is stripped off and an intrinsic epitaxial layer is grown on